

Progressive Retrieval Attention on OpenVINO GenAI: Semantic Resource Identity, Continuous-Batching Evidence, and the Native K/V Boundary

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Abstract

OpenVINO GenAI provides continuous batching, bounded K/V scheduling, prefix caching, eviction controls, and execution across Intel CPU, GPU, and NPU devices. Progressive Retrieval Attention (PRA) adds semantic, query-addressed resources that must remain distinct from sequential prefixes. We implement an OpenVINO provider, an E0 selected-text adapter, and tested contracts for topology-bound native K/V handles, authorization, request pinning, exactly-once attachment, cleanup, and prefix-pool exclusion. A live audit of OpenVINO 2026.3.1 and OpenVINO GenAI 2026.3.1.0 confirms all required continuous-batching controls, but finds no public operation for attaching arbitrary selected non-prefix K/V. We therefore report native E2 as blocked, not as an implicit consequence of prefix caching. On Qwen2-0.5B-Instruct INT4, selected-text and full-context E0 recover the controlled answer in every evidence-bearing request on an Intel i7-1355U CPU and Iris Xe GPU. Selected text uses 63 input tokens instead of 1,573. On GPU it reduces cold TTFT from 1,129.6 to 102.9 ms. Under explicitly warmed continuous batching at eight shared requests, it reaches 8.48 requests/s rather than 7.39 and reduces TTFT p99 from 297.7 to 148.9 ms. An expanded ten-wave sweep reaches concurrency 16: selected context retains 100% controlled recovery and 188.9 output tokens/s, whereas shared full context follows a decoy in every request and reaches 141.1 output tokens/s. A separate 360-request natural cohort covers QASPER, HotpotQA, and 2WikiMultiHopQA; on Hotpot, selected context raises token F1 from 0.080 to 0.310 while cutting TTFT p99 from 2.50 s for full context to 387 ms. A three-point pressure curve keeps selection at 63 tokens while full context grows from 417 to 831 tokens; selected recovery remains exact throughout. The result establishes a reproducible transport baseline and a precise extension boundary; it does not establish native OpenVINO PRA parity or memory savings.

1 Qualification Snapshot

Engine question. Does PRA selection improve quality-adjusted Intel serving beyond OpenVINO’s prefix cache, and is detached native K/V available? The first answer is positive for the measured E0 workloads; the second remains blocked by the public attachment surface.

Table 1: Standard PRA engine result block. NOT MEASURED denotes a missing measurement, not zero cost or zero benefit.

Field	Result
Validated / observed level	E0 validated on CPU and GPU; E2 and E3 not observed
Model / hardware	Qwen2-0.5B-Instruct INT4; i7-1355U and Iris Xe
Quality	Controlled success 1.00; E0 selected natural token F1 0.078–0.310
Visible context	63 selected tokens versus 417–1,573 full-context tokens
Loaded serving	11.1–14.2 selected requests/s and 233–330 ms TTFT p99 at concurrency 16
Native K/V tokens and bytes	NOT MEASURED: no public non-prefix attachment operation
Transfer / E2 reuse / E3 scheduling	NOT MEASURED
Evidence tier	Controlled CPU/GPU transport plus a 360-request natural-QA cohort

Recommendation. Use E0 selected text when frozen retrieval materially removes distractors, and evaluate successful tasks or output-token throughput beside raw requests/s. On this small natural cohort, selected evidence improves HotpotQA F1 and latency, but full context remains stronger on some datasets. Do not claim E2 until OpenVINO exposes selected non-prefix K/V to attention and a selector-frozen E0/E2 comparison establishes parity and reuse economics.

Primer. E0 renders the selected evidence as ordinary prompt text. E1 preserves PRA resource identities, typed records, routing, and fallback policy through the runtime while attention remains ordinary. E2 exposes detached selected native K/V directly to attention. E3 additionally makes their physical placement, sharing, promotion, and eviction scheduler-owned. This paper validates E0 and specifies, but does not measure, the native levels.

2 Introduction

An inference engine’s prefix cache identifies previously computed sequential token histories. PRA identifies immutable logical resources selected by a query. These identities have different lifetimes and authorization rules:

$$I_{\text{prefix}} \neq I_{\text{PRA}} \neq I_{\text{physical}}.$$

Identical visible prompts may select different protected resources, while the same resource may be shared safely by many sessions. A correct OpenVINO integration must preserve that distinction while allowing OpenVINO to keep control of compiled execution, K/V blocks, eviction, and batching.

This study makes three contributions. It implements the PRA-side OpenVINO contracts, audits the installed GenAI scheduler and pipeline interfaces, and measures a common E0 workload across CPU, GPU, prefix reuse, and continuous batching. The audit also identifies the smallest missing native mechanism: request attention metadata cannot currently receive an arbitrary selected non-prefix K/V set through the public Python API.

3 Claim Boundary

E0 renders selected source text into the visible prompt. E1 carries PRA logical identities, typed records, routing decisions, and explicit fallback policy while model attention remains ordinary. E2 consumes selected detached K/V natively, preserving source geometry and combining local and memory scores in one softmax. E3 gives the engine scheduler physical ownership of native PRA placement, sharing, promotion, eviction, and accounting. All model measurements in this paper are E0. The native classes described below are executable lifecycle contracts, not evidence that OpenVINO already performs E2 or E3.

4 OpenVINO Mapping

4.1 Responsibility split

PRA owns typed records, selection, authorization, task/session policy, semantic retention, and a four-tier storage lifecycle. OpenVINO owns compilation, device execution, sequential K/V blocks, prefix reuse, batching, and physical eviction. Physical tensor handles are derived state bound to a specific model, revision, device, precision, layer count, K/V-head count, and head dimension.

4.2 Implemented code

`src/pras/openvino/adapter.py` provides `OpenVINOEngineAdapter`. Its HTTP/OVMS form is explicitly E0. It reports E2 only when an application supplies an `OpenVINONativeExecutor`; package installation or prefix-cache availability is never enough.

`src/pras/openvino/native.py` provides a topology fingerprint, immutable K/V handles, a physical-store protocol, an in-memory deterministic store, and `OpenVINONativeAttachmentManager`. The manager validates topology before registration, authorizes a complete routed set before scheduler admission, pins it for the request lifetime, returns it exactly once, and unpins it during cleanup. A shared isolation guard rejects any PRA logical key offered to the ordinary sequential/prefix pool. Focused tests cover topology mismatch, cross-tenant access, duplicate attachment, cleanup, and capability honesty.

4.3 Live API boundary

`SchedulerConfig` exposes cache size or K/V-block count, sequence and batched-token limits, prefix caching, dynamic split/fuse, and cache eviction. Supplying this configuration activates continuous batching. The public `LLMPipeline` surface exposes generation, tokenizer/config access, and chat state. It does not expose selected external K/V, non-prefix tensor attachment, or an attention-metadata callback. Prefix caching therefore accelerates repeated visible histories but does not implement PRA E2.

5 Experimental Method

5.1 Environment

The remote Windows host runs Python 3.10.0, OpenVINO 2026.3.1, OpenVINO GenAI 2026.3.1.0, and OpenVINO Tokenizers 2026.3.1.0. Available devices are a 13th Gen Intel Core i7-1355U CPU and Intel Iris Xe integrated GPU; no NPU is available. The model is OpenVINO Qwen2-0.5B-Instruct INT4. Scheduler settings use a 1 GiB cache, eight sequences, 2,048 batched tokens, prefix caching, and dynamic split/fuse. Greedy decoding is capped at 24 new tokens.

5.2 Conditions

The target resource contains `PRA_EVIDENCE_4821`; distractors contain different codes. The five one-shot conditions are no evidence, stable prefix only, selected text, prefix plus selected text, and full retained context. The same evidence is used in selected and full conditions. Each condition is run five times independently on CPU and GPU.

The direct `ContinuousBatchingPipeline` experiment uses selected text and full context under shared-resource and independent-resource workloads. Concurrency is 1, 2, 4, and 8 with five measured waves. Every distinct resource is primed outside the timed interval. Reported p95/p99 values are smoke descriptors over small cohorts, not production service-level estimates. The expanded campaign uses ten waves at concurrency 1, 2, 4, 8, and 16, a 4,096-token batch budget, and six repeated distractors. It records success, requests/s, output tokens/s, TTFT, TPOT, generation time, input tokens, and RSS.

6 Results

6.1 Capability status

Table 2: OpenVINO integration status after live package and hardware audit.

Capability	Status	Evidence
Selected-text E0	Ready	CPU/GPU live generation
Prefix cache controls	Ready	live SchedulerConfig audit
Continuous batching	Ready	direct batched generation
PRA lifecycle/isolation	Contract ready	focused tests
Native selected K/V E2	Blocked	no public attachment operation
OVMS E0	Adapter ready	endpoint not installed on host
NPU comparison	Not run	no NPU device available

6.2 CPU and GPU one-shot behavior

Table 3: Evidence-bearing E0 conditions. Visible is mean input tokens; times are milliseconds. Success is controlled answer recovery.

Device	condition	visible	success	cold TTFT	warm TTFT	latency p50
CPU	pra_only	63	1.00	549.2	336.8	1641.2
CPU	prefix_plus_pra	361	1.00	909.8	191.2	1118.1
CPU	full_context	1573	1.00	7600.3	180.6	1249.1
GPU	pra_only	63	1.00	102.9	99.1	788.3
GPU	prefix_plus_pra	361	1.00	115.0	98.5	814.6
GPU	full_context	1573	1.00	1129.6	98.2	501.3

Controls recover the answer in 0/10 requests across CPU and GPU; every evidence-bearing condition succeeds in 30/30. Selected text uses 63 tokens, 96.0% fewer than the 1,573-token full context. On CPU, cold TTFT falls from 7,600.3 ms for full context to 549.2 ms for selected text. On GPU it falls from 1,129.6 to 102.9 ms. Prefix caching changes the comparison: warm full-context TTFT is 180.6 ms on CPU and 98.2 ms on GPU, versus 336.8 and 99.1 ms for selected text. Thus selected text strongly reduces cold ingestion, while a fully reusable prefix can remove most repeated-prefill cost in this fixed one-resource smoke.

Pipeline construction is reported separately: 1.58 s for the CPU one-shot process and 39.40 s for the first GPU one-shot process. The later direct continuous-batching GPU process constructs in 5.05 s after system/model caches have been populated; these are cold-process observations, not comparable steady-state request latencies.

6.3 Continuous batching

With one shared resource and concurrency eight, selected text reaches 8.48 requests/s compared with 7.39 for full context, while TTFT p99 falls from 297.7 to 148.9 ms. For independent resources, selected text still lowers TTFT p99 (262.0 versus 319.8 ms) but reaches only 5.88 requests/s versus 7.07. This small reversal prevents a blanket throughput claim. It may reflect output length, scheduler packing, and churn in the explicitly primed independent set; larger cohorts and matched generated-token work are required.

RSS spans roughly 1.95–2.19 GiB across endpoint rows and is not monotonic with transport or concurrency. The current experiment therefore supports a token and latency claim, not an engine-memory saving claim.

Table 4: Explicitly primed GPU continuous-batching endpoints. Every row has controlled answer success 1.00. RSS is process resident memory after the group.

Transport / resources	C	req/s	TTFT p99	generation p99	RSS MiB
selected / shared	1	1.35	159.7	791.6	2140
selected / shared	8	8.48	148.9	957.8	1950
selected / independent	1	1.57	115.6	760.3	1960
selected / independent	8	5.88	262.0	1430.8	2147
full / shared	1	1.55	223.7	832.1	2163
full / shared	8	7.39	297.7	1154.7	2115
full / independent	1	1.46	188.3	850.3	2119
full / independent	8	7.07	319.8	1192.3	2193

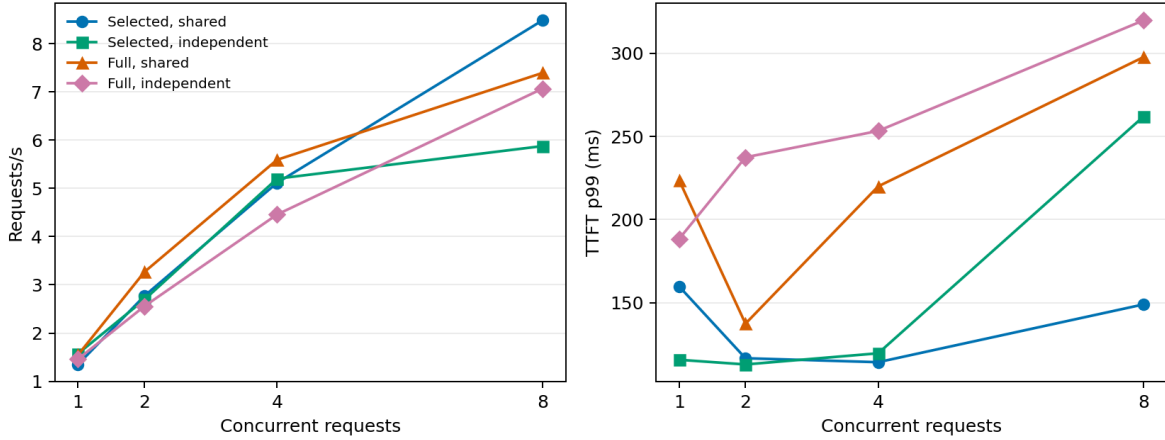


Figure 1: Warm OpenVINO GenAI E0 throughput and TTFT tail under direct continuous batching. Selected denotes rendered selected text, not native E2 K/V.

6.4 Natural QA and expanded continuous batching

Table 5: Natural-QA E0 cohort: 20 examples per dataset and two repeats. Times are milliseconds; retrieval is frozen across the execution conditions.

Dataset	Context	F1	Contain.	Prompt tok.	TTFT p50	TTFT p95
2Wiki	None	0.078	0.150	39	180.5	237.9
2Wiki	Selected	0.121	0.275	387	234.8	389.8
2Wiki	Full	0.137	0.275	877	267.7	1887.8
hotpotqa	None	0.080	0.050	45	179.1	253.3
hotpotqa	Selected	0.310	0.350	428	277.8	378.6
hotpotqa	Full	0.412	0.475	1339	527.0	2432.4
qasper	None	0.066	0.050	34	168.4	284.1
qasper	Selected	0.124	0.050	411	241.4	384.6
qasper	Full	0.219	0.250	1561	705.5	2737.9

The Qwen2-0.5B INT4 checkpoint shows useful but incomplete QA competence. Selected context improves token F1 over no context on all datasets: 0.121 versus 0.078 on 2Wiki, 0.310 versus 0.080 on HotpotQA, and 0.124 versus 0.066 on QASPER. Full context is stronger in this bounded cohort, with F1 0.137, 0.412, and 0.219, exposing a retrieval/evidence-coverage gap rather than a transport failure. Selected median TTFT is 235–278 ms, compared with 268–705 ms for full context. The tail difference is larger: selected p95 is 379–390 ms, whereas full-context p95 is 1.89–2.74 s.

Table 6: Expanded GPU batching endpoints. Each row has ten waves. Success and output-token throughput are shown because wrong short answers can inflate raw requests/s.

Workload	Context	C	Succ.	Req/s	Out tok/s	TTFT p50	TTFT p99	ITL p99
Shared	Selected	1	1.00	1.45	24.7	101.7	126.1	38.2
Shared	Selected	8	1.00	8.33	141.6	137.5	157.7	6.6
Shared	Selected	16	1.00	11.11	188.9	297.7	329.8	5.3
Independent	Selected	1	1.00	1.33	19.4	120.4	148.8	60.8
Independent	Selected	8	1.00	5.38	82.6	231.7	280.6	12.7
Independent	Selected	16	1.00	8.36	124.9	354.7	440.3	7.8
Shared	Full	1	0.00	1.68	16.8	113.9	203.7	69.1
Shared	Full	8	0.00	8.23	82.3	216.0	302.4	11.8
Shared	Full	16	0.00	14.11	141.1	317.6	351.2	5.8
Independent	Full	1	0.30	1.64	17.0	126.1	192.9	70.1
Independent	Full	8	0.51	7.05	75.1	220.8	301.3	13.8
Independent	Full	16	0.45	12.21	129.3	282.3	375.1	6.7

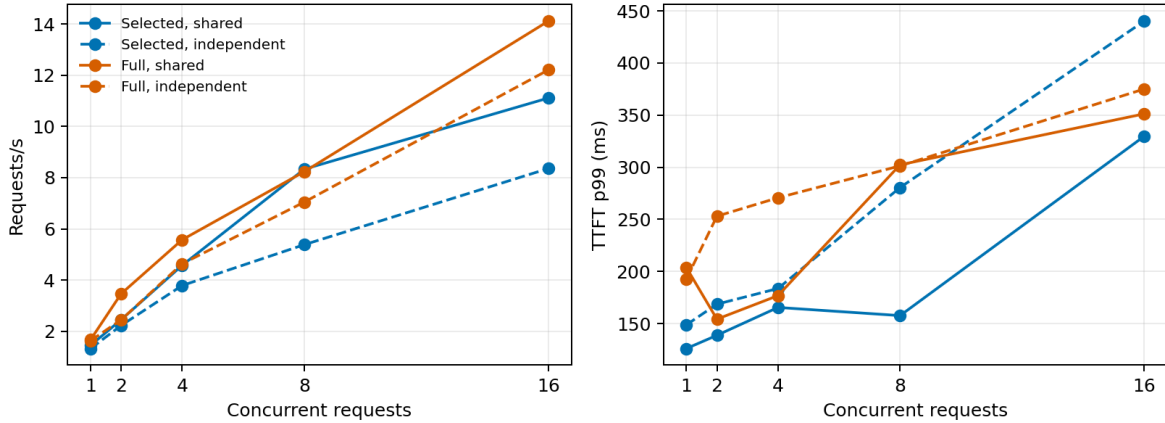


Figure 2: Expanded OpenVINO E0 throughput and TTFT tails through concurrency 16. The full-context request/s lines include wrong, shorter completions and must be read with Table 6.

Selected context recovers the target in every expanded request. At shared concurrency 16 it reaches 11.1 requests/s and 188.9 output tokens/s, compared with 14.1 requests/s and 141.1 output tokens/s for full context. The latter follows a distractor in all 160 requests and terminates after ten tokens rather than 17. Under independent resources, selected remains exact while full context success ranges from 0.30 to 0.51. At concurrency 16, selected reaches 8.36 requests/s, 124.9 output tokens/s, and 440 ms TTFT p99; full reaches 12.2, 129.3, and 375 ms but only 0.45 success. This campaign therefore establishes a quality-adjusted selected-context advantage, not a blanket latency win.

6.5 Context-pressure curve

To separate concurrency from retained-context size, we repeat the peak-load endpoint at three frozen prompt constructions. The selected condition is unchanged at 63 input tokens; the full condition grows from 417 to

677 and 831 tokens. Each point uses concurrency 16 and ten measured waves after explicit warmup.

Table 7: OpenVINO context pressure at concurrency 16. Success is reported beside throughput because the model sometimes follows a distractor and emits a shorter answer.

Size	Workload	Context	Prompt tok.	Succ.	Req/s	Out tok/s	TTFT p99
Small	Shared	Selected	63	1.00	14.16	240.7	233.1
Small	Independent	Selected	63	1.00	12.75	188.1	242.7
Small	Shared	Full	417	1.00	11.14	267.4	168.5
Small	Independent	Full	417	0.88	8.15	153.0	246.5
Medium	Shared	Selected	63	1.00	11.11	188.9	329.8
Medium	Independent	Selected	63	1.00	8.36	124.9	440.3
Medium	Shared	Full	677	0.00	14.11	141.1	351.2
Medium	Independent	Full	677	0.45	12.21	129.3	375.1
Large	Shared	Selected	63	1.00	14.12	240.0	234.8
Large	Independent	Selected	63	1.00	10.81	161.1	318.6
Large	Shared	Full	831	1.00	11.52	126.7	483.3
Large	Independent	Full	831	0.82	10.18	112.6	529.7

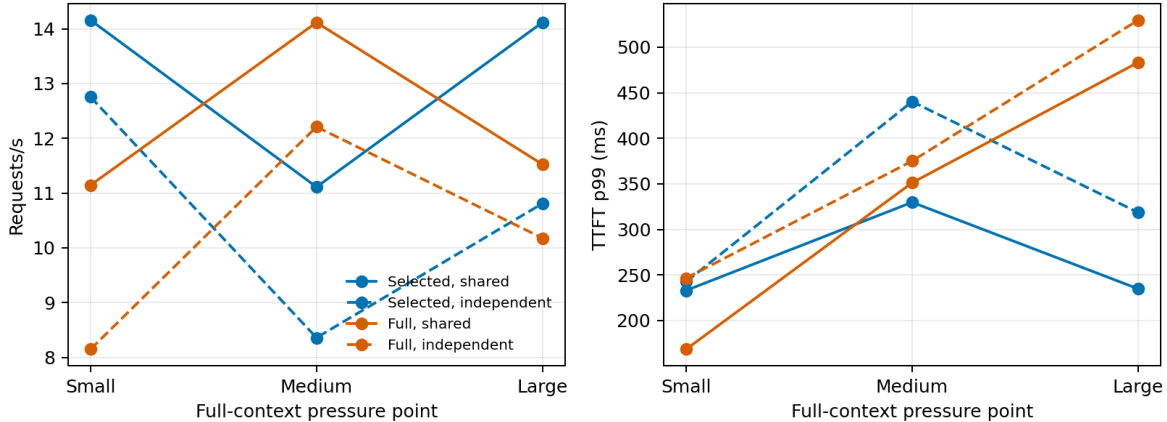


Figure 3: Peak-load throughput and TTFT tail as full visible context grows. Selected context remains fixed; solid lines share one resource and dashed lines use independent resource identities.

Selected evidence remains exact at all three pressure points. Its shared throughput ranges from 11.1 to 14.2 requests/s and TTFT p99 from 233 to 330 ms without increasing visible input. Full-context TTFT p99 grows from 169 ms at 417 tokens to 483 ms at 831 tokens for a shared resource, and from 247 to 530 ms for independent resources. Full-context correctness is not monotonic: it is exact at the smallest and largest shared points but follows a decoy throughout the middle shared point. This is a prompt-order/model effect, not evidence loss, and prevents interpreting raw throughput as a monotonic quality curve. The stable result is narrower: selection bounds input work and preserves the controlled target as retained context grows.

7 Interpretation

OpenVINO prefix caching and PRA selection address different reuse axes. When the same full context is repeatedly visible, prefix caching can make its warm TTFT competitive with selected text. When requests arrive concurrently, shorter selected inputs reduce scheduler work and improve the shared-resource frontier in this smoke. Native PRA would target a harder regime: encode a resource once, select different non-prefix intervals for many queries, and avoid making those source tokens visible. That regime cannot be measured until the attention attachment seam exists.

8 Required Native Extension

A minimal extension should accept an authorized set of topology-matched K/V handles plus source positions at request admission. The scheduler must pin the set, keep it outside ordinary prefix identity, expose it once to attention, and release it on completion or cancellation. Attention must normalize local and memory scores jointly. A geometry ladder should then compare ordinary prefix, geometry-matched cached source, full native source, sparse multi-record source, and cached decode before any economics claim.

9 Limitations

The controlled task verifies evidence transport; the natural cohort adds only 20 examples per dataset. Only one small INT4 model, one CPU, one integrated GPU, and at most ten waves per endpoint are measured. There is no OVMS process, NPU, native K/V parity, device-memory telemetry, long-session pressure, eviction sweep, or larger natural QA cohort. All serving results are E0.

10 Conclusion

PRA can coexist with OpenVINO's public serving abstractions at E0 while preserving semantic identity and explicit capability reporting. Selected text substantially reduces cold ingestion and natural-input TTFT, and it avoids the decoy failures seen in the expanded full-context control. OpenVINO GenAI exposes the physical scheduling controls needed around a future integration but not the public arbitrary non-prefix K/V attachment needed for E2. That boundary is now implemented as a safe contract and measured as an open gate.

Reproducibility

Raw JSON, generated tables and plots, environment audit, and runners are under `docs/papers/shared/results/paper6_3_openvino` and `experiments/paper6_3_openvino`.

References

- [1] Intel. *OpenVINO GenAI LLMPipeline and SchedulerConfig API Reference*. https://docs.openvino.ai/2026/api/genai_api/
- [2] OpenVINO Toolkit. *OpenVINO GenAI*. <https://github.com/openvinotoolkit/openvino.genai>